Interview Assignment: Build an AI-Assisted Software Engineering System - URL Shortener

1. Objective

Build a working prototype that transforms a requirement into a reviewable engineering outcome using AI-assisted engineering execution. Demonstrate requirement understanding, task decomposition, multi-step execution, and output generation/validation. Focus on engineer-led execution accelerated by AI, not autonomous orchestration.

2. Scenario

You will build a URL shortener service from scratch with core APIs, analytics, and reliability features.
Your task is to complete and improve it over 2-3 days using AI assistance (Copilot/Claude/etc.) while **demonstrating engineering judgment**.

3. Scope

- Greenfield scenarios (new systems/features)
- Brownfield scenarios (enhancements, refactors, bug fixes)
- Test and documentation improvements
- Well-defined and ambiguous requirements

4. Core Requirements

1) Requirement Understanding: Interpret intent, identify ambiguity, normalize into a clear engineering problem.

2) Task Decomposition: Convert high-level requirements into actionable tasks with dependencies and sequencing.

3) Codebase Reasoning (Brownfield): Identify impacted modules/services/APIs/data flows and demonstrate architectural understanding.

4) AI-Assisted Execution (Critical Differentiator): Use AI across implementation, debugging, refactoring, test generation, documentation, and review preparation. Define tasks with intent, constraints, acceptance criteria, and technical context; use disciplined prompting with iterative refinement; maintain traceability (generated/edited/rejected with rationale); apply quality gates (analysis, linting, tests, security, performance); enforce secure AI usage;

Classification: Schwab Internal

require human sign-off for high-impact changes; and retain explicit engineer ownership of correctness, maintainability, and production readiness.

5) Engineering Output Generation: Produce production-quality code, API/schema definitions, unit/integration tests, and supporting documentation with clean design and maintainability.

6) Validation and Risk Control: Identify risks/trade-offs/failure scenarios and define validation and safety guardrails.

7) Controlled Oversight: Engineer leads execution and approves all outputs; AI assists within tasks.

8) Final Engineering Summary: Include plan/rationale, artifacts, risks/trade-offs/validation, assumptions, and limitations.

5. Deliverables

- Working prototype (runnable end-to-end)
- Architecture overview (components, tools, execution approach, control flow, key decisions)
- Three scenarios: greenfield, brownfield, ambiguous (each shows decomposition, execution, validation)
- Setup instructions
- Testing approach, limitations, and trade-offs

6. Evaluation Criteria

- Effectiveness of AI-assisted engineering execution
- Architecture/system design quality
- Depth of decomposition and execution quality
- Realism/quality of outputs
- Validation and risk management rigor
- Clarity and defensibility of decisions
- Core engineering principles: modular, testable, reliable, secure, scalable code with safe change management
- Engineering judgment

7. Expectation

Treat as production-grade engineering work. Demonstrate strong design fundamentals, effective AI use as accelerator, output ownership, and defensible reasoning. Principle: AI assists the engineer within tasks; the engineer owns execution and quality.